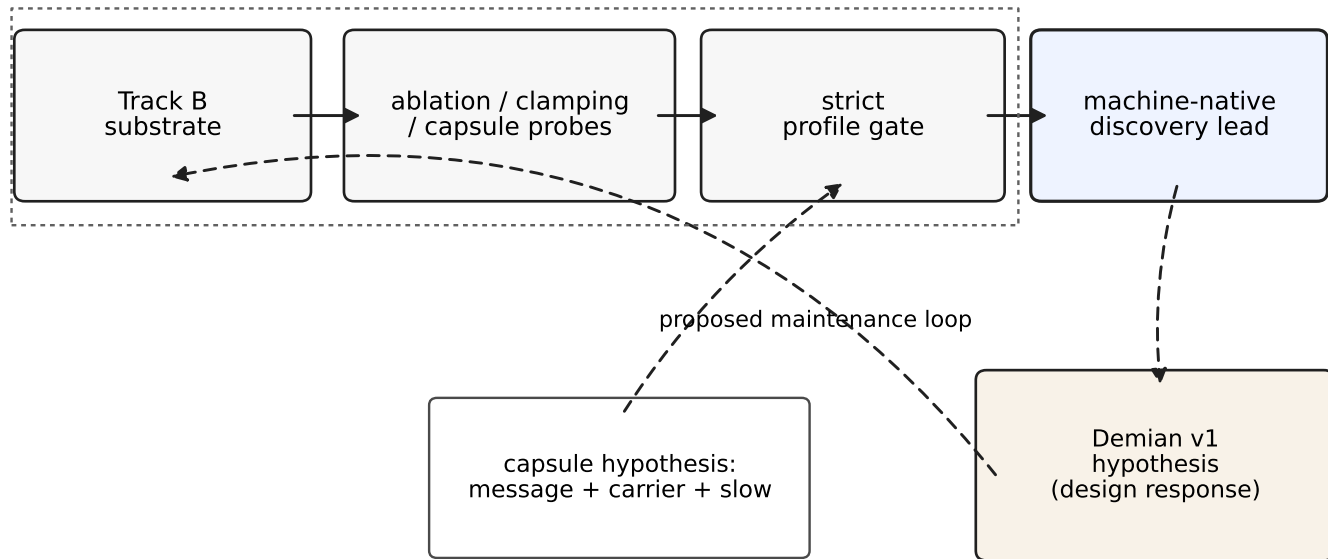


current evidence: five-channel diagnostic protocol



dashed arrows mark proposed maintenance, not control-loop proof